9.7.1

EE25BTECH11057 - Rushil Shanmukha Srinivas

Question : Find the area of the region $\{S(x,y) : x^2 + y^2 \leq 16a^2 \ and \ y^2 \leq 6ax\}$.
Solution :

Name	Value
Parabola	$\mathbf{x}^\top \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x} - \begin{pmatrix} 6a & 0 \end{pmatrix} \mathbf{x} = 0$
Circle	$\mathbf{x}^\top \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x} - 16a^2 = 0$

Table : Conics

On comparing conics with

$$\mathbf{x}^\top \mathbf{V}\mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \tag{0.1}$$

The parameters of the Parabola are

$$\mathbf{V_1} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \mathbf{u_1} = -\begin{pmatrix} 3a \\ 0 \end{pmatrix}, f_1 = 0 \tag{0.2}$$

The parameters of the Circle are

$$\mathbf{V_2} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \mathbf{u_2} = 0, f_2 = -16a^2 \tag{0.3}$$

The intersection of two conics with parameters $\mathbf{V_i}, \mathbf{u_i}, f_i$, i = 1,2 is defined as

$$\mathbf{x}^\top (\mathbf{V_1} + \mu \mathbf{V_2})\mathbf{x} + 2(\mathbf{u_1} + \mu \mathbf{u_2})^\top \mathbf{x} + (f_1 + \mu f_2) = 0. \tag{0.4}$$

(**??**) represents a pair of straight lines if

$$\begin{vmatrix} \mathbf{V} & \mathbf{u} \\ \mathbf{u}^\top & f \end{vmatrix} = 0 \tag{0.5}$$

From (**??**) , (**??**) represents a pair of straight lines if

$$\begin{vmatrix} \mathbf{V_1} + \mu \mathbf{V_2} & \mathbf{u_1} + \mu \mathbf{u_2} \\ (\mathbf{u_1} + \mu \mathbf{u_2})^\top & f_1 + \mu f_2 \end{vmatrix} = 0 \tag{0.6}$$

On substituting from (**??**) and (**??**) in (**??**), we obtain

$$\begin{vmatrix} \mu & 0 & -3a \\ 0 & \mu + 1 & 0 \\ -3a & 0 & -16\mu a^2 \end{vmatrix} = 0 \tag{0.7}$$

On solving we get

$$(\mu + 1)(16\mu^2 + 9) = 0 \tag{0.8}$$

yielding

$$\mu = -1 \tag{0.9}$$

Substituting (**??**) and (**??**) in (**??**), we obtain

$$\mathbf{x}^\top \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix} \mathbf{x} + 2 \begin{pmatrix} -3a & 0 \end{pmatrix} \mathbf{x} + 16a^2 = 0 \tag{0.10}$$

On solving we get

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \mathbf{x} = 2a \ and \ \begin{pmatrix} 1 & 0 \end{pmatrix} \mathbf{x} = -8a \tag{0.11}$$

But circle and line $\begin{pmatrix} 1 & 0 \end{pmatrix} \mathbf{x} = -8a$ do not intersect. So The solution is

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \mathbf{x} = 2a \tag{0.12}$$

Therefore the intersection of two conics is a line with parameters

$$\mathbf{m} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \mathbf{h} = \begin{pmatrix} 2a \\ 0 \end{pmatrix} \tag{0.13}$$

The points of intersection for a conic and a line is given by

$$\kappa_i = \frac{1}{\mathbf{m}^\top \mathbf{V} \mathbf{m}} \left(-\mathbf{m}^\top (\mathbf{V}\mathbf{h} + \mathbf{u}) \pm \sqrt{\left[\mathbf{m}^\top (\mathbf{V}\mathbf{h} + \mathbf{u})\right]^2 - g(\mathbf{h}) \left(\mathbf{m}^\top \mathbf{V} \mathbf{m}\right)} \right) \tag{0.14}$$

$$g(\mathbf{x}) = \mathbf{x}^\top \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x} - 16a^2 \tag{0.15}$$

$$g(\mathbf{h}) = \mathbf{h}^\top \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{h} - 16a^2 \tag{0.16}$$

The intersection parameters of the chord in (**??**) with the circle is obtained as

$$\kappa_i = \pm \sqrt{16a^2 - \mathbf{h}^\top \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{h}} \tag{0.17}$$

$$\kappa_i = \pm 2\sqrt{3}a \tag{0.18}$$

So the points of intersection are obtained as

$$\mathbf{x_1} = \begin{pmatrix} 2a \\ 2\sqrt{3}a \end{pmatrix}, \mathbf{x_2} = \begin{pmatrix} 2a \\ -2\sqrt{3}a \end{pmatrix} \tag{0.19}$$

The desired area of region is given as

$$2 \left(\int_0^{2a} \sqrt{6ax}\, dx + \int_{2a}^{4a} \sqrt{16a^2 - x^2}\, dx \right) \tag{0.20}$$

$$= 2\left[\sqrt{6a}\left(\frac{2}{3}x^{3/2}\right)\right]_0^{2a} + 2\left[\frac{x}{2}\sqrt{16a^2 - x^2} + 8a^2\sin^{-1}\left(\frac{x}{4a}\right)\right]_{2a}^{4a} \tag{0.21}$$

$$= \frac{4a^2}{\sqrt{3}} + \frac{16\pi a^2}{3} \tag{0.22}$$

$$= 4a^2\left(\frac{4\pi}{3} + \frac{1}{\sqrt{3}}\right) \tag{0.23}$$

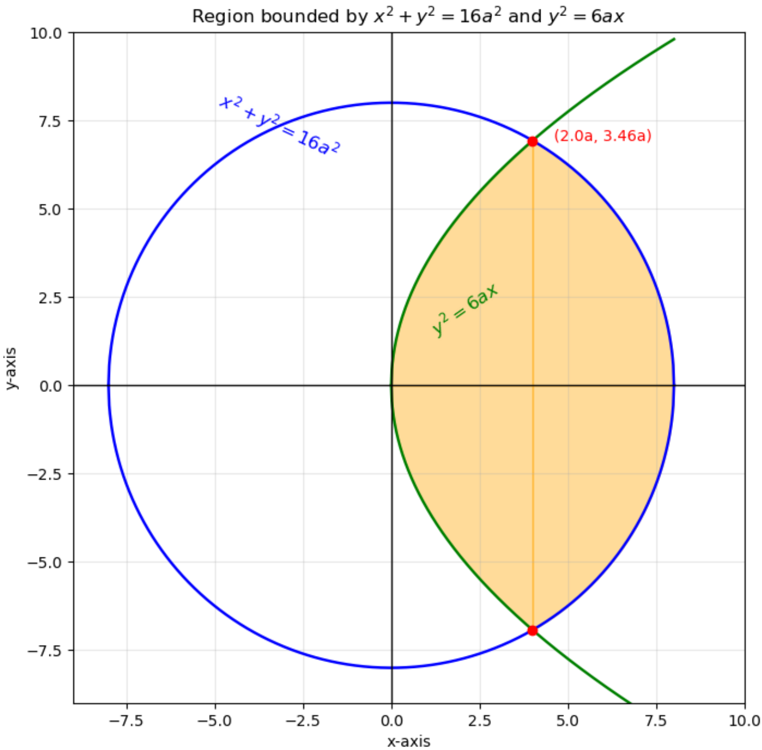

Fig : Parabola and Circle